Sequences And Series

Chapter Overview

Solution 2

1. Arithmetic series

Determine the value of

$2 + 4 + 6 + \cdots + 100$.

2. Geometric series

The first term of a geometric sequence is 3 and the second term is 1. Find its sum to infinity.

3. Sigma notation

Determine the value of

$\sum_{r=1}^{100}(3r + 1)$.

4. Recurrence relations

If $a_1 = k$ and $a_{n+1} = 2a_n - 1$, determine a_3 in terms of k.

Types of sequences

Solution 3

1. $2,\ 5,\ 8,\ 11,\ 14, \ldots$

Name this type of sequence and describe the change between consecutive terms.

1

Arithmetic sequence

Add the same amount each time: $+3$.

The **common difference** is $d = 3$.

2. $3,\ 6,\ 12,\ 24,\ 48, \ldots$

Name this type of sequence and describe the change between consecutive terms.

2

Geometric sequence

Multiply by the same number each time: $\times 2$.

The **common ratio** is $r = 2$.

3. $1,\ 1,\ 2,\ 3,\ 5,\ 8, \ldots$

How is each new term generated?

3

This is the **Fibonacci sequence**. After the first two terms, each term is the sum of the preceding two.

This is a **recurrence relation**: new terms are defined using earlier terms.

Definition

An arithmetic sequence has a constant difference between consecutive terms.

Geometric sequences and recurrence relations will be explored later in this chapter.

The fundamentals of sequences

Solution 4

Sequence notation

Consider

$2,\ 5,\ 8,\ 11,\ 14, \ldots$

What does n represent?

n is the **position** of a term in the sequence.

Positions start at $n = 1$.

What does u_n represent?

u_n is the **value of the term** at position n.

For example, u_3 means the third term.

For the value 8, state its position and its term notation.

$n = 3, \qquad u_3 = 8.$

Do not confuse the position 3 with the value 8.

nth term of an arithmetic sequence

Solution 5

The general term

Let a be the first term and d the common difference.

Moving to position n takes $n-1$ steps.

$u_1 = a,$

$u_2 = a + d,$

$u_3 = a + 2d.$

$$\boxed{u_n = a + (n-1)d}$$

Write down a, d and the relevant position before substituting.

Example 1

$u_n = 55 - 2n.$

(a) Find the first three terms.

(b) Find the first negative term.

1a $u_1 = 53, \quad u_2 = 51, \quad u_3 = 49.$

1b

$55 - 2n < 0 \Rightarrow n > 27.5.$

The first integer position is $n = 28$:

$u_{28} = 55 - 56 = -1.$

Example 2

Find the general term of each sequence.

(a) $6, 20, 34, 48, 62, \ldots$

(b) $101, 94, 87, 80, 73, \ldots$

2a

$a = 6, d = 14$:

$u_n = 6 + 14(n-1) = 14n - 8.$

2b

$a = 101, d = -7$:

$u_n = 101 - 7(n-1) = 108 - 7n.$

Further Examples

Solution 6

1. Find the constants

A sequence is generated by $u_n = an + b$, where a and b are constants. Given $u_3 = 5$ and $u_8 = 20$, find a and b.

1

$3a + b = 5,$

$8a + b = 20.$

Subtracting gives $5a = 15$, so $a = 3$.

Then $3(3) + b = 5$, giving $b = -4$.

$$\boxed{u_n = 3n - 4}$$

Here a is the coefficient of n, not the first term.

2. Equal differences

For which values of x do -8, x^2 and $17x$ form the first three terms of an arithmetic sequence?

2

Consecutive differences must be equal:

$$x^2 - (-8) = 17x - x^2,$$

$$2x^2 - 17x + 8 = 0,$$

$$(2x - 1)(x - 8) = 0.$$

$$\boxed{x = \tfrac{1}{2} \text{ or } x = 8}$$

Each value gives equal differences.

Test Your Understanding — Solution 7

Edexcel C1 May 2014(R), Q10

Xin has a 14-day training schedule. She runs for A minutes on day 1 and increases her running time by $d + 1$ minutes each day, where A and d are constants.

(a) Show that on day 14 she runs for $A + 13d + 13$ minutes. (2 marks)

a

There are 13 increases between days 1 and 14:

$$X_{14} = A + 13(d + 1) = A + 13d + 13.$$

Yi also has a 14-day schedule. She runs for $A - 13$ minutes on day 1 and increases her running time by $2d - 1$ minutes each day.

(b) Yi and Xin run for the same time on day 14. Find d. (3 marks)

b

$$Y_{14} = (A - 13) + 13(2d - 1)$$
$$= A + 26d - 26.$$

Equating the two times:

$$A + 13d + 13 = A + 26d - 26,$$
$$39 = 13d,$$
$$d = 3.$$

Exercise 3A, Y2 pages 61-62 — Solution 8

4. The first term of an arithmetic sequence is 14. The fourth term is 32. Find the common difference.

4

$$u_4 = a + 3d,$$
$$32 = 14 + 3d,$$
$$d = \frac{32 - 14}{3} = 6.$$

5. A sequence is generated by $u_n = pn + q$. Given $u_6 = 9$ and $u_9 = 11$, find p and q.

5

$$6p + q = 9,$$
$$9p + q = 11.$$

Subtracting gives $3p = 2$, so $p = \frac{2}{3}$.

Substitute into the first equation:

$$6\left(\frac{2}{3}\right) + q = 9 \Rightarrow q = 5.$$

$$\boxed{p = \tfrac{2}{3}, \quad q = 5}$$

Exercise 3A, Y2 pages 61-62 — Solution 9

6. An arithmetic sequence has $u_3 = 30$ and $u_9 = 9$. Find its first negative term.

6

$$d = \frac{9-30}{9-3} = -\frac{7}{2}.$$

$$u_n = 30 - \frac{7}{2}(n-3) = \frac{81-7n}{2}.$$

Hence $u_n < 0$ when $n > \frac{81}{7}$.

The first such position is 12:

$$u_{12} = -\frac{3}{2}.$$

7. The 20th term is 14 and the 40th term is -6. Find the 10th term.

7

$$d = \frac{-6-14}{40-20} = -1.$$

Moving back ten positions:

$$u_{10} = u_{20} - 10d = 14 + 10 = 24.$$

8. The first three terms are $5p$, 20 and $3p$. Find the 20th term.

8

$$20 - 5p = 3p - 20,$$
$$8p = 40,$$
$$p = 5.$$

Thus $a = 25$ and $d = -5$:

$$u_{20} = 25 + 19(-5) = -70.$$

Exercise 3A, Y2 pages 61-62 — Solution 10

10. An arithmetic sequence has first term k^2 and common difference k, where $k > 0$. The fifth term is 41. Find k, giving your answer in the form $p + q\sqrt{5}$, where p and q are integers. (4 marks)

10

Using $u_5 = a + 4d$:

$$k^2 + 4k = 41.$$

Complete the square:

$$k^2 + 4k + 4 = 45,$$
$$(k+2)^2 = 45,$$
$$k + 2 = \pm 3\sqrt{5}.$$

Since $k > 0$, reject $-2 - 3\sqrt{5}$.

$$\boxed{k = -2 + 3\sqrt{5}}$$

EXTENSION

Solution 11

STEP I 2004, Q5. The positive integers split into five arithmetic progressions:

$A: 1, 6, 11, 16, \ldots$ $B: 2, 7, 12, 17, \ldots$ $C: 3, 8, 13, 18, \ldots$

$D: 4, 9, 14, 19, \ldots$ $E: 5, 10, 15, 20, \ldots$

Give the general term of each progression. Prove that any term in B plus any term in C belongs to E.

For positive integer positions n:

$A_n = 5n - 4$, $B_n = 5n - 3$,
$C_n = 5n - 2$, $D_n = 5n - 1$,
$E_n = 5n$.

The positions may differ:

$$\begin{aligned} B_n + C_m &= (5n - 3) + (5m - 2) \\ &= 5(n + m - 1) \\ &= E_{n+m-1}. \end{aligned}$$

Since $n, m \geq 1$, the position $n + m - 1$ is positive.

Prove that the square of every term in B belongs to D. State and prove a similar claim for the square of every term in C.

$$\begin{aligned} (5n - 3)^2 &= 25n^2 - 30n + 9 \\ &= 5(5n^2 - 6n + 2) - 1. \end{aligned}$$

Thus B_n^2 belongs to D. Also

$$\begin{aligned} (5n - 2)^2 &= 25n^2 - 20n + 4 \\ &= 5(5n^2 - 4n + 1) - 1. \end{aligned}$$

Thus C_n^2 belongs to D too. Both displayed positions are positive integers for $n \geq 1$.

Show there are no positive integer solutions:

(i) $x^2 + 5y = 243723$.

(ii) $x^4 + 2y^4 = 26081974$.

i

(i) Modulo 5, squares have remainders 0, 1 or 4. Adding $5y$ changes none of these.

The right side has remainder 3, so equality is impossible.

(ii) Modulo 5, fourth powers are 0 or 1. Thus $x^4 + 2y^4$ can only be 0, 1, 2 or 3.

The right side is 4 modulo 5, so equality is impossible.

Series

Solution 12

What is a series?

A series is a sum of terms in a sequence. In this syllabus, a series may be finite or infinite.

Arithmetic series

Add terms of an arithmetic sequence:

$2 + 5 + 8 + 11$.

Consecutive terms have a constant difference.

Taylor series

Functions can be expressed as infinite series of powers of x. For example:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This is studied in Further Mathematics.

Terminology

A power series is an infinite series of increasing non-negative integer powers of a variable, with constant coefficients. Power series also appear in Further Statistics.

Binomial series

A negative or fractional power can produce an infinite expansion. For $|x| < 1$:

$$\sqrt{1 + x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \cdots$$

This is studied later in Year 2.

Partial sums

The harmonic series is

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

A partial sum uses a finite number of terms:

$$H_3 = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}.$$

Arithmetic Series

Solution 13

The sum of an arithmetic series

Let a be the first term, d the common difference and S_n the sum of the first n terms.

Formula

$$S_n = \frac{n}{2}\big(2a + (n-1)d\big)$$

Exam note

Be able to explain this proof. Symmetrically opposite terms add to the same total; the forward-and-reverse argument works for both odd and even n.

See the pairing

Consider $2, 5, 8, 11, 14, 17, \ldots$.
Write the first five terms forwards and backwards.

$S_5 = 2 + 5 + 8 + 11 + 14,$

$S_5 = 14 + 11 + 8 + 5 + 2.$

Add corresponding terms.

Example

Each pair sums to 16:

$$\begin{aligned} 2S_5 &= 16 + 16 + 16 + 16 + 16 \\ &= 5 \times 16 = 80, \\ S_5 &= 40. \end{aligned}$$

Prove the formula

Write the terms forwards and backwards, with $L = a + (n-1)d$. Each corresponding pair has the same sum.

Proof

$S_n = a + (a+d) + \cdots + L,$

$S_n = L + (L-d) + \cdots + a.$

Each pair sums to $a + L$, so

$2S_n = n(a + L).$

Substitute $L = a + (n-1)d$:

$2S_n = n\big(2a + (n-1)d\big).$

Divide by 2 to obtain the formula.

Alternative Formula

Solution 14

Using the first and last terms

Suppose an arithmetic series has n terms, first term a and last term L:

$a + (a+d) + \cdots + L.$

Pair the two copies

Write the terms forwards and backwards. Each pair consists of terms equally far from opposite ends, so each pair sums to $a + L$.

There are n pairs in the two copies together.

Formula

$2S_n = n(a + L),$

$$\boxed{S_n} = \boxed{\frac{n}{2}(a + L)}.$$

The sum is the number of terms multiplied by the mean of the first and last terms.

Connection to the earlier formula

The last term is $L = a + (n-1)d$. Substitute this into the first-and-last-term formula.

Connection

$$\begin{aligned} S_n &= \frac{n}{2}\big(a + a + (n-1)d\big) \\ &= \frac{n}{2}\big(2a + (n-1)d\big). \end{aligned}$$

Both formulas give the same sum. Choose the one suited to the information given.

Examples

Solution 15

Find the sum of the first 30 terms.

Write down a, d and n before substituting. Use S_n to distinguish a sum from the nth term.

1. $2 + 5 + 8 + 11 + 14 + \cdots$

1

$a = 2,\ d = 3,\ n = 30.$

$$S_{30} = \frac{30}{2}(4 + 29 \times 3)$$
$$= 15 \times 91 = 1365.$$

2. $100 + 98 + 96 + \cdots$

2

$a = 100,\ d = -2,\ n = 30.$

$$S_{30} = \frac{30}{2}(200 - 58)$$
$$= 15 \times 142 = 2130.$$

3. $p + 2p + 3p + \cdots$

3

$a = p,\ d = p,\ n = 30.$

$$S_{30} = \frac{30}{2}(2p + 29p)$$
$$= 465p.$$

4. Find the minimum number of terms for $4 + 9 + 14 + \cdots$ to exceed 2000.

4

$a = 4,\ d = 5$, so $S_n = \frac{n}{2}(5n + 3)$. We need $5n^2 + 3n - 4000 > 0$.

The positive root is $\frac{-3+\sqrt{80009}}{10} \approx 27.986$. Since n is a positive integer, $n \geq 28$.

Check: $S_{27} = 1863 < 2000$ and $S_{28} = 2002 > 2000$. The minimum is $\boxed{28}$ terms.

Test Your Understanding

Solution 16

Edexcel C1, January 2012, Q9

A company offers two salary schemes for Years 1 to 10 inclusive. Scheme 1 starts at £P, increasing by £$2T$ each year. Scheme 2 starts at £$(P + 1800)$, increasing by £T each year.

(a) Show that the ten-year total under Scheme 1 is £$(10P + 90T)$. (2 marks)

a

First term P, difference $2T$, ten terms:

$$S_{10}^{(1)} = \frac{10}{2}(2P + 9(2T))$$
$$= 5(2P + 18T)$$
$$= 10P + 90T.$$

(b) Both schemes give the same ten-year total. Find T. (4 marks)

b

For Scheme 2:

$$S_{10}^{(2)} = 5\big(2(P + 1800) + 9T\big)$$
$$= 10P + 18000 + 45T.$$

Equate the totals:

$$90T = 18000 + 45T,$$
$$45T = 18000,$$
$$T = 400.$$

(c) Scheme 2 pays £29,850 in Year 10. For this value of T, find P. (3 marks)

c

The tenth term is the first plus nine increases:

$$P + 1800 + 9T = 29850,$$
$$P + 1800 + 3600 = 29850,$$
$$P = 24450.$$

Scheme 1 therefore starts at £24,450.

Exercise 3B, Year 2 Pages 64-66 — Solution 17

3. Find the sum of the first 50 even numbers.

3

The first 50 positive even numbers are $2, 4, 6, \ldots, 100$.

Here $a = 2,\ d = 2,\ n = 50$.

$$S_{50} = \frac{50}{2}\left(2 \times 2 + 49 \times 2\right)$$
$$= 25 \times 102$$
$$= \boxed{2550}.$$

4. Find the least number of terms for $7 + 12 + 17 + 22 + 27 + \cdots$ to exceed 1000.

4

$a = 7,\ d = 5$, so $S_n = \frac{n}{2}(5n + 9)$.

$S_n > 1000 \iff 5n^2 + 9n - 2000 > 0$.

The positive root is $\frac{-9+\sqrt{40081}}{10} \approx 19.12$. Hence $n \geq 20$.

Check: $S_{19} = 988$ and $S_{20} = 1090$. The least number is $\boxed{20}$.

Exercise 3B, Year 2 Pages 64-66 — Solution 18

5. The first term of an arithmetic series is 4. The sum to 20 terms is -15. Find, in any order, the common difference and the 20th term.

Find the common difference

Difference

Use $a = 4,\ n = 20,\ S_{20} = -15$:

$$-15 = \frac{20}{2}(8 + 19d),$$
$$-1.5 = 8 + 19d,$$
$$19d = -9.5,$$
$$d = \boxed{-0.5}.$$

Find the twentieth term

20th term

$$u_{20} = a + 19d$$
$$= 4 + 19(-0.5)$$
$$= \boxed{-5.5}.$$

Check using the first and last terms:

$$S_{20} = \frac{20}{2}(4 - 5.5) = -15.$$

6. The sum of the first three terms of an arithmetic series is 12. The twentieth term is -32. Find the first term and common difference.

6

$$a + (a + d) + (a + 2d) = 12,$$
$$a + d = 4,$$
$$a + 19d = -32.$$

Subtract the second equation from the third:

$18d = -36 \quad \Rightarrow \quad d = -2.$

Then $a - 2 = 4$, giving $\boxed{a = 6,\ d = -2}$.

7. Prove that the sum of the first 50 natural numbers is 1275.

7

Write the sum in both directions:

$S_{50} = 1 + 2 + \cdots + 49 + 50,$

$S_{50} = 50 + 49 + \cdots + 2 + 1.$

Adding gives 50 pairs, each summing to 51:

$2S_{50} = 50 \times 51,$

$$S_{50} = \frac{50 \times 51}{2} = \boxed{1275}.$$

8. Show that the sum of the first $2n$ natural numbers is $n(2n + 1)$.

8

The series is $1 + 2 + \cdots + 2n$, with $2n$ terms, first term 1 and last term $2n$.

$$S_{2n} = \frac{2n}{2}(1 + 2n)$$
$$= \boxed{n(2n + 1)}.$$

Note that the number of terms here is $2n$, not n.

9. Prove that the sum of the first n odd numbers is n^2.

9

The series is $1 + 3 + 5 + \cdots + (2n - 1)$.

It has n terms, first term 1 and common difference 2.

$$S_n = \frac{n}{2}\big(2 + (n - 1)2\big)$$
$$= \frac{n}{2}(2n)$$
$$= \boxed{n^2}.$$

Exercise 3B, Year 2 Pages 64-66 — Solution 21

10. The fifth term of an arithmetic series is 33 and the tenth term is 68. The sum of the first n terms is 2225.

(a) Show that $7n^2 + 3n - 4450 = 0$.

(b) Hence find n.

a

$a + 4d = 33,$

$a + 9d = 68.$

Subtract to get $5d = 35$, so $d = 7$ and $a = 5$.

$$2225 = \frac{n}{2}\big(10 + 7(n-1)\big),$$

$$4450 = n(7n + 3),$$

$$7n^2 + 3n - 4450 = 0.$$

b

$$n = \frac{-3 \pm \sqrt{9 + 4 \times 7 \times 4450}}{14}$$

$$= \frac{-3 \pm 353}{14}.$$

Thus $n = 25$ or $n = -\frac{178}{7}$.

The number of terms must be positive, so $\boxed{n = 25}$.

Exercise 3B, Year 2 Pages 64-66 — Solution 22

11. An arithmetic series is $(k+1) + (2k+3) + (3k+5) + \cdots + 303$.

(a) Find the number of terms in terms of k.

(b) Show that the sum is $\frac{152k+46208}{k+2}$.

(c) Given $S_n = 2568$, find k.

a

$a = k + 1$, $d = k + 2$. The last term is 303:

$$303 = k + 1 + (n-1)(k+2)$$

$$= n(k+2) - 1,$$

$$n = \frac{304}{k+2}.$$

b

Use the first and last terms:

$$S_n = \frac{n}{2}(a + 303)$$

$$= \frac{152}{k+2}(k + 304)$$

$$= \frac{152k + 46208}{k+2}.$$

c

$$2568(k+2) = 152k + 46208,$$

$$2416k = 41072,$$

$$k = \boxed{17}.$$

Check: $n = 304/19 = 16$, and $S_{16} = 8(18 + 303) = 2568$.

EXTENSION

1. MAT 2008, 1I

For positive integers n, let $S(n)$ be the sum of its digits; for example, $S(723) = 12$. Find

$$S(1) + S(2) + \cdots + S(99).$$

1

Include 00, whose digit sum is zero. Among 00–99, every digit occurs ten times in each position.

Units contribute $10(1 + 2 + \cdots + 9) = 450$. Tens contribute another 450.

Total $= 450 + 450 = \boxed{900}$.

2. AEA 2010, Q2

An arithmetic series satisfies $S_p = q$ and $S_q = p$, where p, q are positive integers and $p \neq q$.

In terms of p, q, find:

(a) the common difference;

(b) the first term;

(c) S_{p+q}.

Full solution on the next slide.

3. MAT 2007, 1J

For every $n \geq 1$,

$$(n+1) + (n^4 + 2) + \cdots + (n^{10000} + 100) > k.$$

Which follows?

A: $k < 1300$ B: $k^2 < 101$

C: $k \geq 101^{10000}$

D: $k < 5150$

3

The left side is smallest at $n = 1$. Its minimum is

$$100 + \frac{100 \times 101}{2} = 5150.$$

Therefore $k < 5150$: $\boxed{\text{D}}$.

Solution to Extension Q2

AEA 2010, Q2

An arithmetic series has $S_p = q$ and $S_q = p$, where $p, q > 0$ are distinct integers. Find its common difference d, first term a and S_{p+q}.

(a) Common difference

a

$$\frac{2q}{p} = 2a + (p-1)d,$$

$$\frac{2p}{q} = 2a + (q-1)d.$$

Subtract:

$$(p-q)d = \frac{2(q^2 - p^2)}{pq},$$

$$d = -\frac{2(p+q)}{pq}.$$

Cancellation is valid since $p \neq q$.

(b) First term

b

From the first sum equation:

$$2a = \frac{2q}{p} - (p-1)d.$$

Substitute d:

$$a = \frac{q}{p} + \frac{(p-1)(p+q)}{pq}$$

$$= \frac{p^2 + pq + q^2 - p - q}{pq}.$$

(c) Sum to $p+q$ terms

c

$$S_{p+q} = \frac{p+q}{2}\big(2a + (p+q-1)d\big).$$

Substituting the values from (a) and (b), the bracket simplifies to

$$\frac{-2pq}{pq} = -2.$$

Therefore

$$\boxed{S_{p+q} = -(p+q)}.$$

Recap of Arithmetic vs Geometric Sequences

Solution 25

Compare the two sequences

Identify the type of sequence and its constant step.

1. $2, 5, 8, 11, 14, \ldots$

1

An **arithmetic sequence**.

Add 3 each time:

$d = 5 - 2 = 8 - 5 = \boxed{3}$.

The constant step is the **common difference**, d.

2. $3, 6, 12, 24, 48, \ldots$

2

A **geometric sequence**.

Multiply by 2 each time:

$r = \dfrac{6}{3} = \dfrac{12}{6} = \boxed{2}$.

The constant multiplier is the **common ratio**, r.

Definition

A geometric sequence is one in which each term is obtained by multiplying the previous term by the same constant ratio.

Quickfire Common Ratio

Solution 26

Identify the common ratio r.

Divide a term by the preceding non-zero term.

1. $1, 2, 4, 8, 16, 32, \ldots$

1

$r = 2/1 = 2$

2. $27, 18, 12, 8, \ldots$

2

$r = 18/27 = \dfrac{2}{3}$

3. $10, 5, 2.5, 1.25, \ldots$

3

$r = 5/10 = \dfrac{1}{2}$

4. $5, -5, 5, -5, 5, -5, \ldots$

4

$r = -5/5 = -1$

5. $x, -2x^2, 4x^3, \ldots$

5

$r = -2x \quad (x \neq 0)$

6. $1, p, p^2, p^3, \ldots$

6

$r = p$

7. $4, -1, 0.25, -0.0625, \ldots$

7

$r = -1/4 = -\dfrac{1}{4}$

Alternating signs

A negative common ratio makes non-zero terms alternate between positive and negative. For item 5, if $x = 0$, all displayed terms are zero and the ratio cannot be uniquely determined.

nth term

Solution 27

Arithmetic sequence

Arithmetic formula

$u_n = a + (n-1)d.$

1. Find the tenth and nth terms of $3, 6, 12, 24, \ldots$.

1

$a = 3,\ r = 2.$

$u_{10} = 3 \times 2^9 = 1536,$

$u_n = 3 \times 2^{n-1}.$

Geometric sequence

Geometric formula

$u_n = ar^{n-1}.$

2. Find the tenth and nth terms of $40, -20, 10, -5, \ldots$.

2

$a = 40,\ r = -\frac{1}{2}.$

$u_{10} = 40\left(-\frac{1}{2}\right)^9 = -\frac{5}{64},$

$u_n = 40\left(-\frac{1}{2}\right)^{n-1}.$

Equivalently, $u_n = (-1)^{n-1}\frac{5}{2^{n-4}}.$

Tip

Write down a and r before substituting. The exponent is $n-1$, because there are $n-1$ multiplications after the first term.

Further Example

Solution 28

Textbook example

The second term of a geometric sequence is 4 and the fourth term is 8. Its common ratio is positive. Find the exact values of the common ratio, the first term and the tenth term.

(a) Common ratio

a

$u_2 = 4 \Rightarrow ar = 4,$

$u_4 = 8 \Rightarrow ar^3 = 8.$

Divide the second equation by the first:

$r^2 = 2 \quad \Rightarrow \quad \boxed{r = \sqrt{2}}.$

Choose the positive root as stated.

(b) First term

b

$a = \frac{4}{r}$

$= \frac{4}{\sqrt{2}}$

$= \boxed{2\sqrt{2}}.$

(c) Tenth term

c

$u_{10} = ar^9$

$= (2\sqrt{2})(\sqrt{2})^9$

$= 2(\sqrt{2})^{10}$

$= \boxed{64}.$

Tip

Writing $u_2 = 4$ first helps distinguish the second term from the sum of the first two terms.

Further Example

Solution 29

The first three terms of a geometric sequence are $3, x, x+6$. All terms are positive.

Find x and the tenth term.

1. Find x

1

Consecutive ratios are equal:

$$\frac{x}{3} = \frac{x+6}{x}.$$

$$x^2 = 3x + 18,$$

$$x^2 - 3x - 18 = 0,$$

$$(x-6)(x+3) = 0.$$

Thus $x = 6$ or $x = -3$. All terms are positive, so $\boxed{x = 6}$.

2. Find the tenth term

2

$$a = 3, \qquad r = \frac{6}{3} = 2.$$

Use $u_n = ar^{n-1}$:

$$u_{10} = 3 \times 2^9$$

$$= \boxed{1536}.$$

Method

Equate consecutive ratios when three consecutive terms are given. Check which solutions satisfy the conditions in the question.

nth term with inequalities

Solution 30

Find the first term of the geometric sequence $3, 6, 12, 24, \ldots$ that exceeds one million.

1. Find its position

1

$a = 3$, $r = 2$, so $u_n = 3 \times 2^{n-1}$.

$$3 \times 2^{n-1} > 10^6,$$

$$2^{n-1} > \frac{10^6}{3},$$

$$n - 1 > \frac{\log(10^6/3)}{\log 2},$$

$$n > 19.3466\ldots$$

The smallest integer satisfying this is $\boxed{n = 20}$.

2. Calculate and check the term

2

$u_{20} = 3 \times 2^{19} = \boxed{1\,572\,864}$.

The preceding term is

$u_{19} = 3 \times 2^{18} = 786\,432$.

Therefore the twentieth term is the first to exceed one million.

Tip

The position n must be an integer. Use the inequality to choose the first valid position, then calculate the term.

Test Your Understanding — Solution 31

1. A geometric sequence has only positive terms. Its third term is 20 and its fifth term is 80. Find the twentieth term.

1

$ar^2 = 20, \qquad ar^4 = 80.$

Divide to obtain $r^2 = 4$. Positive terms give $r = 2$, then $a = 20/4 = 5$.

$u_{20} = 5 \times 2^{19} = \boxed{2\,621\,440}$.

2. The second, third and fourth terms of a geometric sequence are $x, x+6, 5x-6$.

(a) Find the possible values of x.

(b) Find r, given that it is positive.

(c) Hence find the first term.

2a

(a) Equal ratios give $(x+6)^2 = x(5x-6)$.

$2x^2 - 9x - 18 = 0,$

$(2x+3)(x-6) = 0.$

So $\boxed{x = -\frac{3}{2} \text{ or } 6}$.

2b–c

(b) For $x = 6$, $r = 12/6 = 2$. For $x = -\frac{3}{2}$, $r = -3$.

Since $r > 0$, choose $\boxed{r = 2}$.

(c) The second term is $ar = 6$, so $\boxed{a = 3}$.

Without the positive-ratio condition, the other branch would give $a = \frac{1}{2}$.

Exercise 3C , Year 2 Pages 69-70 — Solution 32

Exercise 3C, question 3

If $3, x, 9$ are the first three terms of a geometric sequence, find:

(a) the exact value(s) of x;

(b) the exact value(s) of the fourth term.

(a) Second term

a

Consecutive ratios are equal:

$\frac{x}{3} = \frac{9}{x},$

$x^2 = 27,$

$x = \boxed{\pm 3\sqrt{3}}$.

Both signs give a geometric sequence.

(b) Fourth term

b

$r = \frac{x}{3} = \pm\sqrt{3}.$

Multiply the third term by the common ratio:

$u_4 = 9r = \boxed{\pm 9\sqrt{3}}$.

The signs match: positive x gives positive u_4, and negative x gives negative u_4.

Sign check

No positive-ratio condition is stated. Remember to consider both square roots when solving an equation of the form $x^2 = k$, where $k > 0$.

Exercise 3C , Year 2 Pages 69-70 — Solution 33

5. The nth term of a geometric sequence is 2×5^n. Find the first and fifth terms.

5

Substitute $n = 1$ and $n = 5$:

$u_1 = 2 \times 5^1 = \boxed{10}$,

$u_5 = 2 \times 5^5 = \boxed{6250}$.

Notice that the exponent is n in the given formula.

6. The sixth term of a geometric sequence is 32 and the third term is 4. Find the first term and the common ratio.

6

$ar^5 = 32, \qquad ar^2 = 4.$

Divide the two equations:

$$r^3 = \frac{32}{4} = 8 \quad \Rightarrow \quad \boxed{r = 2}.$$

Substitute into $ar^2 = 4$:

$a \times 2^2 = 4 \quad \Rightarrow \quad \boxed{a = 1}$.

Exercise 3C , Year 2 Pages 69-70 — Solution 34

8. The first three terms of a geometric sequence are $8 - x, 2x, x^2$, where $x > 0$.

(a) Show that $x^3 - 4x^2 = 0$. (2 marks)

a

Equate consecutive ratios:

$$\frac{2x}{8 - x} = \frac{x^2}{2x}.$$

Cross-multiply:

$$4x^2 = x^2(8 - x),$$
$$4x^2 = 8x^2 - x^3,$$
$$x^3 - 4x^2 = 0.$$

(b) Find the twentieth term. (3 marks)

b

$x^2(x - 4) = 0$.

Since $x > 0$, $x = 4$. The sequence begins $4, 8, 16, \ldots$, so $a = 4$, $r = 2$.

$u_{20} = 4 \times 2^{19}$

$= \boxed{2\,097\,152}$.

(c) Is 4096 a term of the sequence? Give a reason. (1 mark)

c

Solve $u_n = 4096$:

$4 \times 2^{n-1} = 4096,$

$2^{n-1} = 1024 = 2^{10},$

$n = 11.$

Yes: $\boxed{4096 = u_{11}}$, and 11 is a positive integer.

Exercise 3C , Year 2 Pages 69-70 — Solution 35

9. A geometric sequence has first term 200 and common ratio $p > 0$. Its sixth term is 40.

(a) Show that $5\log p + \log 5 = 0$.

(b) Find p to 3 significant figures.

a

(a) $u_6 = 200p^5 = 40$, so $p^5 = \frac{1}{5}$.

$$5\log p = \log\left(\frac{1}{5}\right) = -\log 5,$$

$5\log p + \log 5 = 0$.

(b) $p = 5^{-1/5} = 0.724779\ldots$, so

$\boxed{p = 0.725}$ (3 s.f.).

11. The first three terms of a geometric sequence are $9, 36, 144$. State, with a reason, whether 383616 is a term of the sequence.

11

$a = 9$ and $r = 4$, so $u_n = 9 \times 4^{n-1}$.

The neighbouring terms are

$u_8 = 9 \times 4^7 = 147456$,

$u_9 = 9 \times 4^8 = 589824$.

Since $147456 < 383616 < 589824$, it lies strictly between consecutive terms.

No, 383616 is not a term of this increasing sequence.

Sum of the first n terms of a geometric series — Solution 36

Arithmetic series

Arithmetic formula

$$S_n = \frac{n}{2}\big(2a + (n-1)d\big).$$

Geometric series

Geometric formula

$$S_n = \frac{a(1-r^n)}{1-r} \quad (r \neq 1).$$

If $r = 1$, every term is a, so $S_n = na$.

Proof of the geometric-series formula

Write the sum, multiply by r, then subtract.

Proof

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1},$$

$$rS_n = ar + ar^2 + ar^3 + \cdots + ar^n.$$

All the shared terms cancel:

$$S_n - rS_n = a - ar^n,$$

$$(1-r)S_n = a(1-r^n),$$

$$S_n = \frac{a(1-r^n)}{1-r} \quad (r \neq 1).$$

Examples

Solution 37

Find the sum of the first ten terms of each geometric series.

Use $S_n = \dfrac{a(1-r^n)}{1-r}$ for $r \neq 1$.

1. $3 + 6 + 12 + 24 + 48 + \cdots$

1

$a = 3, r = 2, n = 10.$

$$S_{10} = \frac{3(1-2^{10})}{1-2}$$
$$= 3(1024-1)$$
$$= \boxed{3069}.$$

2. $4 + 2 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$

2

$a = 4, r = \frac{1}{2}, n = 10.$

$$S_{10} = \frac{4\left(1-\left(\frac{1}{2}\right)^{10}\right)}{1-\frac{1}{2}}$$
$$= 8\left(1 - \frac{1}{1024}\right)$$
$$= \boxed{\frac{1023}{128}}.$$

Check

A term u_n and a sum S_n are different quantities. Identify which one the question asks for before choosing a formula.

Harder Example

Solution 38

Find the least value of n such that the sum of $1 + 2 + 4 + 8 + \cdots$ to n terms exceeds $2\,000\,000$.

1. Form and solve an inequality

1

$a = 1, r = 2,$ so

$$S_n = \frac{1-2^n}{1-2} = 2^n - 1.$$

Hence

$$2^n - 1 > 2\,000\,000,$$
$$2^n > 2\,000\,001,$$
$$n > \frac{\log(2\,000\,001)}{\log 2}$$
$$= 20.9315\ldots$$

The least integer is $\boxed{n = 21}$.

2. Verify the boundary

2

$$S_{20} = 2^{20} - 1$$
$$= 1\,048\,575,$$
$$S_{21} = 2^{21} - 1$$
$$= 2\,097\,151.$$

The sum is still below two million after 20 terms, and exceeds it after 21 terms.

Why logarithms?

Taking logarithms lets us solve for an unknown exponent. Since the base 2 exceeds 1, the inequality keeps its direction.

Test Your Understanding — Solution 39

Edexcel C2 June 2011, Q6

The second and third terms of a geometric series are 192 and 144 respectively.

Find (a) the common ratio (2 marks), (b) the first term (2 marks), and (d) the smallest value of n for which the sum of the first n terms exceeds 1000 (4 marks).

a

$$r = \frac{u_3}{u_2} = \frac{144}{192} = \frac{3}{4}.$$

b

$$ar = 192 \Rightarrow a = \frac{192}{3/4} = 256.$$

d

$$S_n = \frac{256(1-(3/4)^n)}{1-3/4} = 1024(1-(3/4)^n).$$

$$S_n > 1000 \Rightarrow (3/4)^n < \frac{24}{1024} = \frac{3}{128},$$

$$n\log(3/4) < \log(3/128).$$

Since $\log(3/4) < 0$, reverse the inequality:

$$n > \frac{\log(3/128)}{\log(3/4)} \approx 13.047.$$

The smallest integer is $\boxed{n = 14}$.

Exercise 3D, Pure 2 Pages 72-73 — Solution 40

4. The sum of the first three terms of a geometric series is 30.5. If the first term is 8, find the possible values of r.

4

$$8 + 8r + 8r^2 = 30.5,$$

$$16r^2 + 16r - 45 = 0,$$

$$(4r-5)(4r+9) = 0.$$

$$\boxed{r = \frac{5}{4} \quad \text{or} \quad r = -\frac{9}{4}}$$

Both values satisfy the original sum.

5. Find the least value of n such that $3 + 6 + 12 + 24 + \cdots$, to n terms, exceeds 1.5 million.

5

Here $a = 3$ and $r = 2$.

$$S_n = \frac{3(2^n-1)}{2-1} = 3(2^n-1).$$

$$3(2^n-1) > 1500000,$$

$$2^n > 500001,$$

$$n > \frac{\log 500001}{\log 2} \approx 18.932.$$

Hence $\boxed{n = 19}$.

Exercise 3D, Pure 2 Pages 72-73 — Solution 41

6. Find the least value of n such that the sum

$5 + 4.5 + 4.05 + \cdots$

to n terms exceeds 45.

Set up the sum

The first term is $a = 5$ and the common ratio is $r = 0.9$.

6

$$S_n = \frac{5(1 - 0.9^n)}{1 - 0.9} = 50(1 - 0.9^n).$$

$$50(1 - 0.9^n) > 45,$$

$$0.9^n < 0.1,$$

$$n \log 0.9 < \log 0.1.$$

Divide by the negative number $\log 0.9$:

$$n > \frac{\log 0.1}{\log 0.9} \approx 21.854.$$

Therefore the least value is $\boxed{n = 22}$.

Exercise 3D, Pure 2 Pages 72-73 — Solution 42

7. A geometric series has first term 25 and common ratio $\frac{3}{5}$. Its sum to k terms is greater than 61.

(a) Show that $k > \frac{\log(0.024)}{\log(0.6)}$.

(b) Find the smallest possible value of k.

7a

$$\frac{25(1 - 0.6^k)}{0.4} > 61,$$

$$1 - 0.6^k > 0.976,$$

$$0.6^k < 0.024.$$

Taking logs and dividing by $\log 0.6 < 0$:

$$\boxed{k > \frac{\log 0.024}{\log 0.6}}.$$

7b

$k > 7.301\ldots$, so the smallest integer is $\boxed{k = 8}$.

Extension — MAT 2010, 1B

The sum of the first $2n$ terms of

$1,\ 1,\ 2,\ \frac{1}{2},\ 4,\ \frac{1}{4},\ 8,\ \frac{1}{8}, \ldots$

is which expression?

A: $2^n + 1 - 2^{1-n}$

B: $2^n + 2^{-n}$

C: $2^{2n} - 2^{3-2n}$

D: $\frac{2^n - 2^{-n}}{3}$

Extension

Split into two interleaved geometric sequences, each with n terms:

$$1 + 2 + \cdots + 2^{n-1} = 2^n - 1,$$

$$1 + \tfrac{1}{2} + \cdots + 2^{1-n} = 2 - 2^{1-n}.$$

Add the sums:

$\boxed{2^n + 1 - 2^{1-n}}$ Answer A.

Divergent vs Convergent

Solution 43

What happens to each running total?

A series converges if its partial sums approach a finite limit. Otherwise it diverges.

1. $1 + 2 + 4 + 8 + 16 + \cdots$

1 **Divergent.** The partial sums grow without bound.

2. $1 - 2 + 3 - 4 + 5 - 6 + \cdots$

2 **Divergent.** Odd partial sums are $1, 2, 3, \ldots$; even partial sums are $-1, -2, -3, \ldots$. They do not approach one limit.

3. $1 + 0.5 + 0.25 + 0.125 + \cdots$

3 **Convergent.** The partial sums approach 2.

Beyond the A level syllabus — for interest

4. The harmonic series:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

4 **Divergent.** Its partial sums grow without bound, despite the individual terms tending to zero.

5. The Basel problem:

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots$$

5 **Convergent.** The sum is $\frac{\pi^2}{6}$.

Sum to Infinity

Solution 44

Compare two geometric series

$1 + 0.5 + 0.25 + \cdots$ converges to 2, whereas $1 + 2 + 4 + \cdots$ grows without bound.

A geometric series with non-zero first term converges when

$$\boxed{|r| < 1} \quad \text{or} \quad -1 < r < 1.$$

What happens to r^n as $n \to \infty$ when $|r| < 1$?

$r^n \longrightarrow 0$.

For example, $(\frac{1}{2})^{100000}$ is extremely close to zero. Negative ratios with $|r| < 1$ also tend to zero, alternating in sign.

Derive the sum to infinity

Start with the finite sum:

$$S_n = \frac{a(1 - r^n)}{1 - r}.$$

Use the limit of r^n to obtain S_∞.

As $n \to \infty$, $r^n \to 0$, so

$$S_\infty = \frac{a(1 - 0)}{1 - r}.$$

$$\boxed{S_\infty = \frac{a}{1 - r} \quad \text{for } |r| < 1}$$

Check the convergence condition before using this formula.

Quickfire Examples

Solution 45

For each geometric series, state the first term a, common ratio r and sum to infinity. Check $|r| < 1$.

1. $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$

1a First term: $a = 1$.

1r Common ratio: $r = \frac{1}{2}$.

1 sum

$$S_\infty = \frac{1}{1 - 1/2} = 2.$$

2. $27, -9, 3, -1, \ldots$

2a First term: $a = 27$.

2r Common ratio: r = -1/3.

2 sum

$$S_\infty = \frac{27}{1 + 1/3} = \frac{81}{4}.$$

3. $p, p^2, p^3, p^4, \ldots$, where $-1 < p < 1$

3a First term: $a = p$.

3r Common ratio: $r = p$.

3 sum

$$S_\infty = \frac{p}{1 - p}.$$

4. $p, 1, \frac{1}{p}, \frac{1}{p^2}, \ldots$. State when the series converges.

4a First term: $a = p$.

4r Common ratio: $r = \frac{1}{p}$.

4 sum

$$S_\infty = \frac{p^2}{p - 1}.$$

Converges when $|p| > 1$.

Further Examples

Solution 46

1. The fourth term of a geometric series is 1.08 and the seventh term is 0.23328.

(a) Show that the series is convergent.

(b) Find its sum to infinity.

1a

$ar^3 = 1.08, \qquad ar^6 = 0.23328.$

Divide the second equation by the first:

$$r^3 = \frac{0.23328}{1.08} = 0.216 \Rightarrow r = 0.6.$$

Since $|r| < 1$, the series converges.

1b

$$a = \frac{1.08}{0.6^3} = 5.$$

$$S_\infty = \frac{5}{1 - 0.6} = 12.5.$$

2. A geometric series has first term a, common ratio r, $S_4 = 15$ and $S_\infty = 16$.

(a) Find the possible values of r.

(b) All terms are positive. Find a.

2a

$$\frac{a(1 - r^4)}{1 - r} = 15, \qquad \frac{a}{1 - r} = 16.$$

Substituting gives

$$16(1 - r^4) = 15 \Rightarrow r^4 = \frac{1}{16}.$$

$$r = \pm\frac{1}{2}$$

2b

All terms are positive, so $r = \frac{1}{2}$.

$$a = 16(1 - r) = 16\left(1 - \frac{1}{2}\right) = 8.$$

Remember: the exponent is n in the sum formula, but $n - 1$ in $u_n = ar^{n-1}$.

Test Your Understanding

Solution 47

Edexcel C2 May 2011, Q6

The second and third terms of a geometric series are 192 and 144 respectively. Find:

(a) the common ratio (2 marks);

(b) the first term (2 marks);

(c) the sum to infinity (2 marks);

(d) the smallest n for which the sum of the first n terms exceeds 1000 (4 marks).

a

$$r = \frac{144}{192} = \frac{3}{4}.$$

b

$$a = \frac{192}{3/4} = 256.$$

c

$|r| < 1$, so

$$S_\infty = \frac{256}{1 - 3/4} = 1024.$$

d

$$1024\big(1 - (3/4)^n\big) > 1000,$$
$$(3/4)^n < \frac{3}{128},$$
$$n\log(3/4) < \log(3/128).$$

Dividing by $\log(3/4) < 0$ reverses the inequality:

$$n > \frac{\log(3/128)}{\log(3/4)} \approx 13.047.$$

Hence the smallest value is $\boxed{n = 14}$.

Exercise 3E, Pure Y2 Pages 75-76

Solution 48

2. A geometric series has first term 10 and sum to infinity 30. Find the common ratio.

2

$$\frac{10}{1 - r} = 30,$$
$$10 = 30(1 - r),$$
$$30r = 20,$$
$$r = \frac{2}{3}.$$

Check: $|\frac{2}{3}| < 1$, so the series converges.

3. A geometric series has first term -5 and sum to infinity -3. Find the common ratio.

3

$$\frac{-5}{1 - r} = -3,$$
$$-5 = -3 + 3r,$$
$$3r = -2,$$
$$r = -\frac{2}{3}.$$

Check: $|-\frac{2}{3}| < 1$, so the series converges.

Exercise 3E, Pure Y2 Pages 75-76

Solution 49

4. A geometric series has sum to infinity 60 and common ratio $\frac{2}{3}$. Find its first term.

4

$$S_\infty = \frac{a}{1-r} \Rightarrow a = S_\infty(1-r).$$

$$a = 60\left(1 - \frac{2}{3}\right) = \boxed{20}.$$

5. A geometric series has common ratio $-\frac{1}{3}$ and sum to infinity 10. Find its first term.

5

$$a = S_\infty(1-r).$$

$$a = 10\left(1 + \frac{1}{3}\right) = \boxed{\frac{40}{3}}.$$

Exercise 3E, Pure Y2 Pages 75-76

Solution 50

6. Find the fraction equal to the recurring decimal $0.\overline{23} = 0.232323\ldots$

Use a geometric series

Write each repeated block as a separate term, then identify a and r.

6

$$0.\overline{23} = \frac{23}{100} + \frac{23}{10000} + \frac{23}{1000000} + \cdots$$

This is geometric with

$$a = \frac{23}{100}, \qquad r = \frac{1}{100}.$$

Since $|r| < 1$:

$$S_\infty = \frac{23/100}{1 - 1/100} = \frac{23/100}{99/100} = \boxed{\frac{23}{99}}.$$

Exercise 3E, Pure Y2 Pages 75-76 Solution 51

1. MAT 2006, 1H

How many solutions does

$2 = \sin x + \sin^2 x + \sin^3 x + \cdots$

have for $0 \le x < 2\pi$?

1

The series has $a = r = \sin x$ and requires $|\sin x| < 1$.

$$2 = \frac{\sin x}{1 - \sin x},$$

$$2 - 2\sin x = \sin x,$$

$$\sin x = \frac{2}{3}.$$

There are **two solutions**, in quadrants I and II. Both satisfy the convergence condition.

2. MAT 2003, 1F

Two players take turns to throw a fair six-sided die until one scores a six. Find the probability that the first player is the first to score a six.

2

The first player can win on their first, second, third, ... turn:

$$P = \sum_{k=0}^{\infty} \frac{1}{6}\left(\frac{25}{36}\right)^k.$$

Here $a = \frac{1}{6}$, $r = \frac{25}{36}$:

$$P = \frac{1/6}{1 - 25/36} = \boxed{\frac{6}{11}}.$$

3. Extension

Determine

$x = 1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \frac{5}{16} + \cdots$

Hint: multiply, then subtract as in the geometric-sum proof.

3

Let S_N be the first N terms. Multiply by 2 and subtract:

$$S_N = 2 + \sum_{k=0}^{N-2} 2^{-k} - \frac{N}{2^{N-1}}.$$

Summing the geometric part gives

$$S_N = 4 - \frac{N+2}{2^{N-1}}.$$

As $N \to \infty$, the final term tends to zero. Therefore

$\boxed{x = 4}$.

Sigma Notation Solution 52

Read sigma notation

$$\sum_{r=1}^{5}(2r+1)$$

The symbol

The Greek capital letter Σ means **sum**.

The limits

Start at $r = 1$, finish at $r = 5$, increasing r by 1 each time.

The expression

Evaluate $2r + 1$ for each value of r, then add the results.

Expansion

$$\sum_{r=1}^{5}(2r+1) = 3 + 5 + 7 + 9 + 11$$

$$= 35.$$

The terms correspond to $r = 1, 2, 3, 4, 5$ respectively.

Recognise the series

A linear expression in the summation variable gives an arithmetic series. Establish its first term a, common difference d and number of terms before applying the sum formula.

Determining the value

Solution 53

1. $\sum_{n=1}^{7} 3n$

1

First terms: $3 + 6 + 9 + \cdots$.

$a = 3,\ d = 3$; there are 7 terms.

$S_7 = \frac{7}{2}(6 + 6 \times 3) = 84.$

2. $\sum_{k=5}^{15} (10 - 2k)$

2

First terms: $0 - 2 - 4 - \cdots$.

$a = 0,\ d = -2$; there are $15 - 5 + 1 = 11$ terms.

$S_{11} = \frac{11}{2}(0 + 10(-2)) = -110.$

3. $\sum_{k=1}^{12} 5 \times 3^{k-1}$

3

First terms: $5 + 15 + 45 + \cdots$.

$a = 5,\ r = 3$; there are 12 terms.

$S_{12} = \frac{5(3^{12} - 1)}{3 - 1} = 1328600.$

4. $\sum_{k=5}^{12} 5 \times 3^{k-1}$

Either start with the $k = 5$ term, or subtract the first four terms from the first twelve.

4 sum

$$\sum_{k=5}^{12} 5 \times 3^{k-1} = S_{12} - S_4$$

$$= 1328600 - \frac{5(3^4 - 1)}{2}$$

$$= 1328400.$$

Testing Your Understanding

Solution 54

Solomon Paper A

Evaluate

$\sum_{r=10}^{30} (7 + 2r).$

(4 marks)

Method 1: Direct

Identify the first term, last term and number of terms.

Method 1

$27 + 29 + 31 + \cdots + 67.$

Here $a = 27$, $d = 2$, and there are $30 - 10 + 1 = 21$ terms.

$S_{21} = \frac{21}{2}(54 + 20 \times 2) = \boxed{987}.$

Method 2: Subtraction

Sum from 1 to 30, then subtract the terms from 1 to 9.

Method 2

The series starting at $r = 1$ is $9 + 11 + 13 + \cdots$.

$S_{30} = \frac{30}{2}(18 + 29 \times 2) = 1140,$

$S_9 = \frac{9}{2}(18 + 8 \times 2) = 153.$

$\sum_{r=10}^{30} (7 + 2r) = 1140 - 153 = \boxed{987}.$

On your calculator — Solution 55

Use the summation function

The ClassWiz and Casio Silver calculators have a Σ function. On the CG50, use **MATH**, then the next-menu arrow, then $\Sigma($.

Enter the expression, summation variable, lower limit and upper limit in the calculator's summation template.

Try it

$$\sum_{k=5}^{12} 2 \times 3^k$$

Expression: 2×3^k

Variable: k

Lower limit: 5

Upper limit: 12

Calculator result

$$\sum_{k=5}^{12} 2 \times 3^k = 1594080.$$

Check the result

The first term is $2 \times 3^5 = 486$, the common ratio is 3, and there are $12 - 5 + 1 = 8$ terms.

Formula check

$$S_8 = \frac{486(3^8 - 1)}{3 - 1} = 1594080.$$

The calculator and series formula agree.

Exercise 3F, Pure 2, Pages 77-78 — Solution 56

6. Show that

$$\sum_{r=1}^{n} 2r = n + n^2.$$

6

$$\sum_{r=1}^{n} 2r = 2 + 4 + 6 + \cdots + 2n.$$

There are n terms, with first term 2 and last term $2n$:

$$S_n = \frac{n}{2}(2 + 2n)$$
$$= n + n^2.$$

7. Show that

$$\sum_{r=1}^{n} 2r - \sum_{r=1}^{n} (2r - 1) = n.$$

7

From question 6, the first sum is $n + n^2$.

The second sum is $1 + 3 + \cdots + (2n - 1)$:

$$\sum_{r=1}^{n} (2r - 1) = \frac{n}{2}(1 + 2n - 1) = n^2.$$

Subtracting:

$$(n + n^2) - n^2 = \boxed{n}.$$

Exercise 3F, Pure 2, Pages 77-78 — Solution 57

9. Find

$$\sum_{r=20}^{\infty} 200\left(\frac{1}{4}\right)^r.$$

9

The first term is $200(1/4)^{20}$ and the ratio is $1/4$.

$$S_\infty = \frac{200(1/4)^{20}}{1-1/4}$$
$$= \frac{800}{3\times 4^{20}}$$
$$= \boxed{\frac{25}{103079215104}}.$$

10. Given

$$\sum_{r=1}^{k}(8+3r) = 377,$$

(a) show that $(3k+58)(k-13)=0$;

(b) hence find k.

10a

The terms are $11, 14, 17, \ldots, 8+3k$.

$$\frac{k}{2}(11+8+3k) = 377,$$
$$3k^2+19k-754=0,$$
$$(3k+58)(k-13)=0.$$

10b

k is a positive integer, so

$\boxed{k=13}$.

Recurrence Relations — Solution 58

Two ways to define a sequence

Position-to-term rule

$u_n = 2n^2+3$

Substitute the position n to find any term directly.

Position-to-term

$u_1 = 5, \quad u_2 = 11, \quad u_3 = 21.$

Term-to-term rule

$u_{n+1} = 2u_n + 4, \qquad u_1 = 3$

Double the previous term, then add 4. This is a **recurrence relation**.

Recurrence

$$u_2 = 2(3)+4 = 10,$$
$$u_3 = 2(10)+4 = 24,$$
$$u_4 = 2(24)+4 = 52.$$

An initial value is essential.

A recurrence describes how to move between terms. The starting value determines which sequence it generates.

Example

Solution 59

Edexcel C1 May 2013 (R), Q6

$x_1 = 1, \qquad x_{n+1} = x_n^2 - kx_n \quad (n \geq 1).$

Here k is constant. (a) Find x_2. (b) Show that $x_3 = 1 - 3k + 2k^2$.

Given $x_3 = 1$, (c) find k; (d) find $\sum_{n=1}^{100} x_n$.

a–b

(a)

$x_2 = 1^2 - k(1) = \boxed{1 - k}$.

(b)

$x_3 = (1-k)^2 - k(1-k)$

$= 1 - 3k + 2k^2$.

c

(c)

$1 - 3k + 2k^2 = 1,$

$k(2k - 3) = 0.$

Thus $\boxed{k = 0 \text{ or } k = 3/2}$. The stated question does not exclude zero.

d

(d) If $k = 3/2$, the terms repeat $1, -1/2$, so the sum is $50(1 - 1/2) = \boxed{25}$.

If $k = 0$, every term is 1 and the sum is $\boxed{100}$.

Recurrences need not be arithmetic or geometric. Look for a repeating pattern before choosing a sum formula.

Test Your Understanding

Solution 60

Edexcel C1 January 2012, Q4

$x_1 = 1, \qquad x_{n+1} = ax_n + 5 \quad (n \geq 1),$

where a is constant.

(a) Write down x_2. (b) Show that $x_3 = a^2 + 5a + 5$.

Given $x_3 = 41$, (c) find the possible values of a.

a–b

(a)

$x_2 = a(1) + 5 = \boxed{a + 5}$.

(b)

$x_3 = a(a+5) + 5$

$= a^2 + 5a + 5$.

c

(c)

$a^2 + 5a + 5 = 41,$

$a^2 + 5a - 36 = 0,$

$(a + 9)(a - 4) = 0.$

$\boxed{a = -9 \text{ or } a = 4}$.

Exercise 3G, Pure 2, Pages 80-81 — Solution 61

1. AEA 2011, Q3

$u_1 = k$, $u_{2n} = pu_{2n-1}$, $u_{2n+1} = qu_{2n}$, for $n \geq 1$.

(a) Write the first six terms.

(b) Show that, for $pq \neq 1$,

$$\sum_{r=1}^{2n} u_r = \frac{k(1+p)(1-(pq)^n)}{1-pq}.$$

(c) With $[x]$ the integer part of x, find

$$\sum_{r=1}^{\infty} 6\left(\frac{4}{3}\right)^{[r/2]}\left(\frac{3}{5}\right)^{[(r-1)/2]}.$$

a

(a)
$k, kp, kpq, kp^2q, kp^2q^2, kp^3q^2$.

(b) Pair adjacent terms: n pairs, first pair $k(1+p)$, ratio pq. Apply the geometric sum.

(c) $k = 6, p = 4/3, q = 3/5$:

$$S_\infty = \frac{6(1+4/3)}{1-4/5} = \boxed{70}.$$

2. MAT 2014, 1H

$F(1) = 0$. For each $n \geq 2$, add to $F(n-1)$:

2 if $2 \mid n$ but $3 \nmid n$;

3 if $3 \mid n$ but $2 \nmid n$;

4 if both divide n;

0 if neither divides n.

Find $F(6000)$.

2

In every block of six integers, the increments total

$0 + 2 + 3 + 2 + 0 + 4 = 11$.

There are 1000 such blocks. The increment at 1 is zero, consistent with $F(1) = 0$.

$\boxed{F(6000) = 11000}$.

3. MAT 2016, 1G

$$x_0 = 1, \qquad x_n = \sum_{k=0}^{n-1} x_k$$

for $n \geq 1$. Determine

$$\sum_{k=0}^{\infty} \frac{1}{x_k}.$$

Work out the first few terms.

Full worked solution on the next slide.

Solution to Extension Question 3 — Solution 62

MAT 2016: Sum of previous terms

$$x_0 = 1, \qquad x_n = \sum_{k=0}^{n-1} x_k \quad (n \geq 1).$$

Find $\displaystyle\sum_{k=0}^{\infty} \frac{1}{x_k}$.

Pattern

Start with the first terms

$x_0 = 1, \quad x_1 = 1, \quad x_2 = 2, \quad x_3 = 4, \quad x_4 = 8.$

For $n \geq 1$,

$$x_{n+1} = \sum_{k=0}^{n} x_k = x_n + x_n = 2x_n.$$

Hence $x_n = 2^{n-1}$ for $n \geq 1$.

Sum

$$\sum_{k=0}^{\infty} \frac{1}{x_k} = 1 + \left(1 + \frac{1}{2} + \frac{1}{4} + \cdots\right) = 1 + \frac{1}{1-1/2} = \boxed{3}.$$

Increasing, decreasing and periodic sequences

Solution 63

Strictly increasing

$u_{n+1} > u_n$ for every n.

Strictly decreasing

$u_{n+1} < u_n$ for every n.

Periodic

$u_{n+k} = u_n$. The least positive such k is the order.

State whether each sequence is increasing, decreasing or periodic. Give the order if periodic.

(a)

$u_{n+1} = u_n + 3, \quad u_1 = 7.$

a

$7, 10, 13, \ldots$

Strictly increasing: each term is 3 greater.

(b)

$u_{n+1} = u_n^2, \quad u_1 = \frac{1}{2}.$

b

$\frac{1}{2}, \frac{1}{4}, \frac{1}{16}, \frac{1}{256}, \ldots$

Strictly decreasing: squaring a number between 0 and 1 makes it smaller.

(c)

$u_{n+1} = \sin(90n^\circ).$

c

Generated terms from u_2:

$1, 0, -1, 0, 1, 0, \ldots$

They repeat with order 4.

Taking $u_1 = 0$ extends this pattern to the whole sequence.

Terminology: "strictly" excludes equal consecutive terms. A constant sequence is neither strictly increasing nor strictly decreasing.

Exercise 3H, Pure 2, Pages 82-83

Solution 64

Exercise 3H, Q3

$u_{n+1} = ku_n, \qquad u_1 = 5.$

Find the range of values of k for which the sequence is strictly decreasing.

Answer

$u_1 = 5, \quad u_2 = 5k, \quad u_3 = 5k^2, \ldots$

If $0 < k < 1$, every term remains positive and is smaller than its predecessor.

Thus $\boxed{0 < k < 1}$.

Why no other values

Check the excluded cases

$k = 0$: $5, 0, 0, \ldots$ is not strictly decreasing.

$k = 1$: the sequence is constant.

$k > 1$: the sequence increases.

$k < 0$: signs alternate, so it cannot decrease at every step.

Exercise 3H, Pure 2, Pages 82-83 — Solution 65

Exercise 3H, Q5

$a_n = \cos(90n^\circ), \qquad n \geq 1.$

(a) Find the order of the sequence.

(b) Find $\sum_{r=1}^{444} a_r$.

a

(a)

$a_1 = 0, \quad a_2 = -1, \quad a_3 = 0, \quad a_4 = 1, \quad a_5 = 0.$

The block $0, -1, 0, 1$ repeats.

The order is $\boxed{4}$.

b

(b) There are $444/4 = 111$ complete blocks.

$\sum_{r=1}^{444} a_r = 111(0 - 1 + 0 + 1) = \boxed{0}.$

Modelling — Solution 66

Bruce models his first-year profit as £20,000, rising by £5,000 each year until it reaches £100,000, then remaining constant.

(a) Find his total profit over 20 years. (b) Give a limitation of this model.

(c) Instead, model profit as increasing by 5% each year. Find the 20-year total.

a

(a) The cap is reached in year n, where

$20000 + 5000(n-1) = 100000 \Rightarrow n = 17.$

The first 17 years total

$S_{17} = \frac{17}{2}(20000 + 100000) = 1020000.$

Add three years at £100,000:

$\boxed{\text{Total} = £1320000}$.

b

(b) Real profit need not rise by the same amount each year, or stay constant after reaching a fixed cap.

c

(c) With $a = 20000$, $r = 1.05$:

$S_{20} = \frac{20000(1.05^{20} - 1)}{0.05}.$

$\boxed{S_{20} \approx £661319.08}$.

Geometric Modelling Example

Solution 67

A sheet of paper is initially 0.5 mm thick. Each fold doubles its thickness.

(a) Find its thickness after 4 folds.

(b) Find its thickness after 20 folds.

(c) Comment on whether this is realistic for a sheet of A4 paper.

a

Model

After n folds, thickness is $T_n = 0.5 \times 2^n$ mm.

(a) $T_4 = 0.5 \times 16 = \boxed{8 \text{ mm}}$.

b

(b)

$T_{20} = 0.5 \times 2^{20} = 524288$ mm.

This is $\boxed{524.288 \text{ m}}$.

c

(c) The doubling model is mathematical. A finite A4 sheet cannot physically be folded 20 times: its shrinking dimensions and growing thickness prevent it.

Test Your Understanding

Solution 68

Edexcel C2 January 2013, Q3

A company predicts profit of £120,000 in 2013, rising by 5% each year.

(a) Show that the predicted 2016 profit is £138,915.

(b) Find the first year with yearly profit exceeding £200,000.

(c) Find total predicted profit from 2013 to 2023 inclusive, to the nearest pound.

a

(a) 2016 is three years after 2013:

$120000(1.05)^3 = \boxed{£138915}$.

b

(b) Let t be years after 2013.

$120000(1.05)^t > 200000$,

$$t > \frac{\log(5/3)}{\log 1.05} \approx 10.47.$$

The first whole value is $t = 11$, giving $\boxed{2024}$.

c

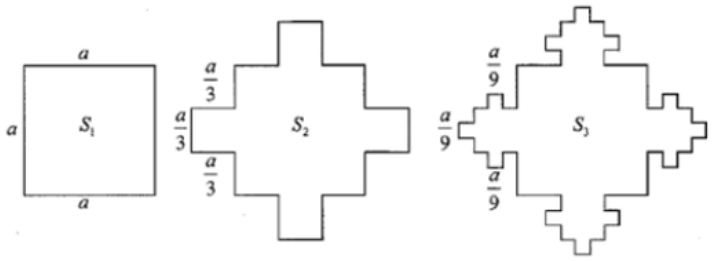
(c) There are $2023 - 2013 + 1 = 11$ years.

$$S_{11} = \frac{120000(1.05^{11} - 1)}{0.05} \approx 1704814.459.$$

To the nearest pound: $\boxed{£1704814}$.

AEA 2007, Q5: Model snowflakes

S_1 is a square of side a. For S_2, replace the middle third of each edge with a new square of side $a/3$. At each later stage, repeat this on each external edge of the squares added at the previous stage, using squares one-third as large.



(a) Show that S_4 adds 36 squares of side $a/27$.

(b) Show that $P_2 = 20a/3$ and $P_3 = 28a/3$.

(c) Find the perimeter P_n.

a

(a) Each new square has three external edges. The numbers added are $4, 12, 36$, while their side lengths are $a/3, a/9, a/27$.

b

(b) Adding a square of side s replaces one edge of length s by three: the net increase is $2s$.

$$P_2 = 4a + 4(2a/3) = 20a/3,$$
$$P_3 = 20a/3 + 12(2a/9)$$
$$= 28a/3.$$

c

(c) Each stage increases perimeter by $8a/3$:

$$P_n = 4a + (n-1)\frac{8a}{3} = \boxed{\frac{(8n+4)a}{3}}.$$